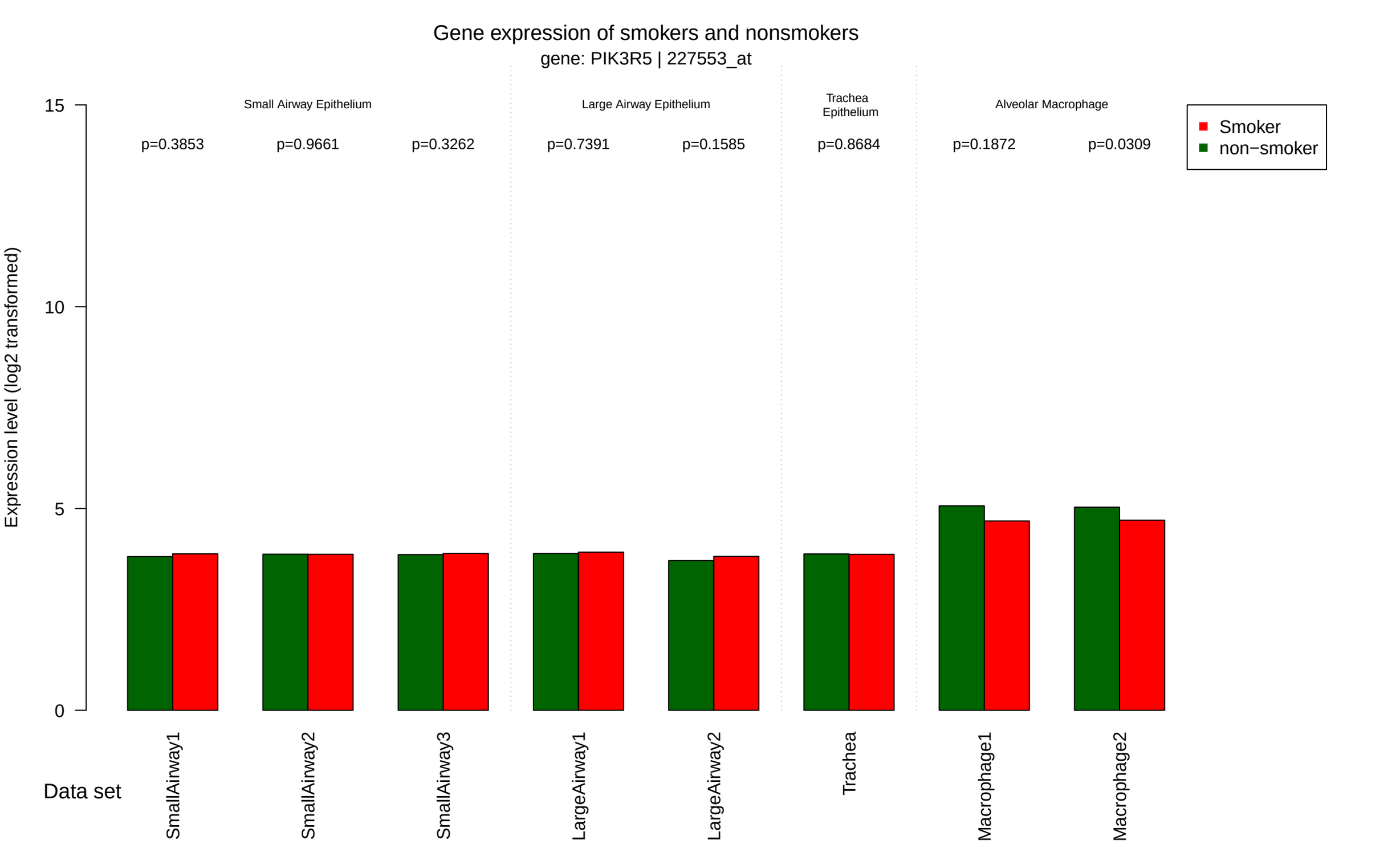
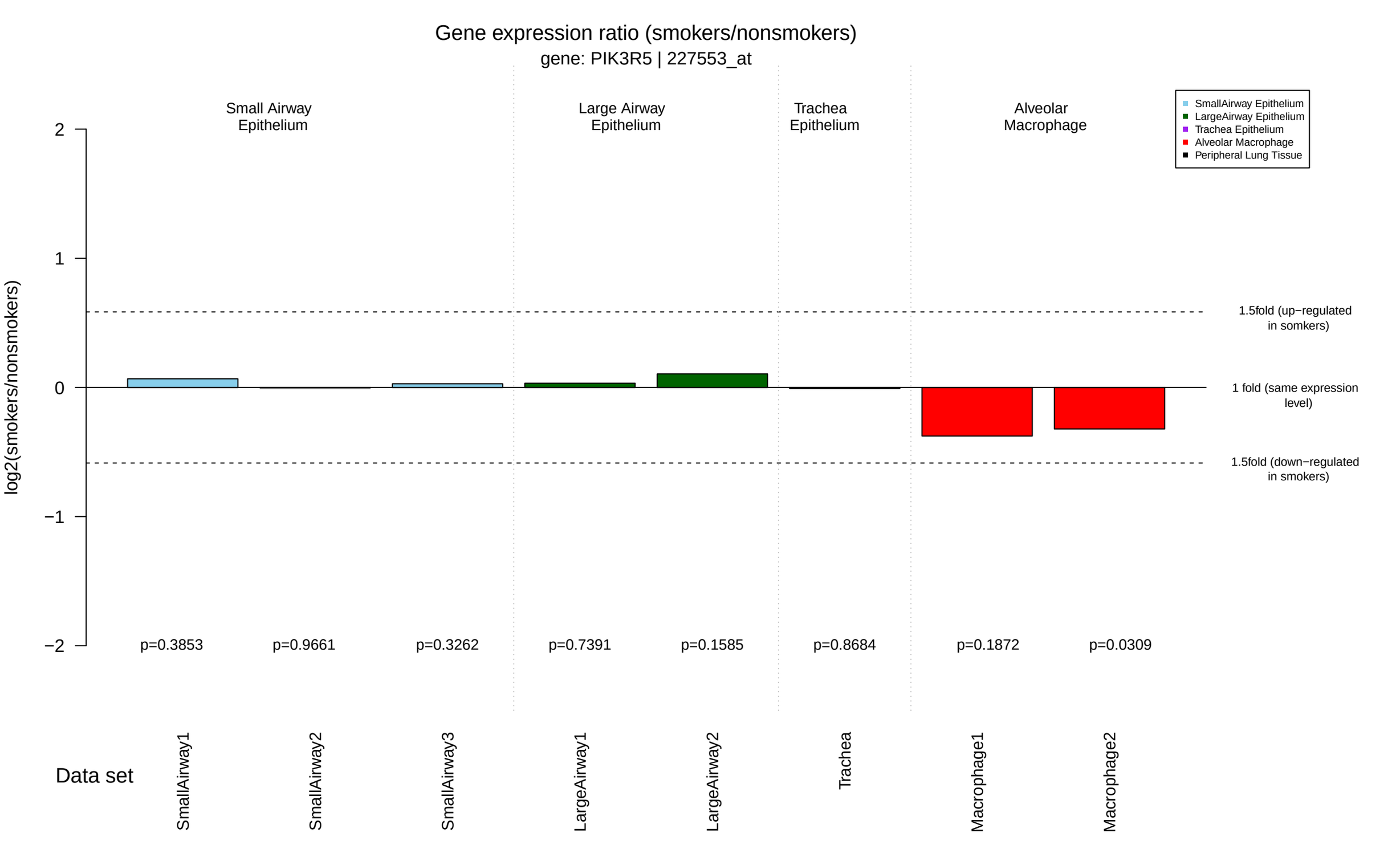
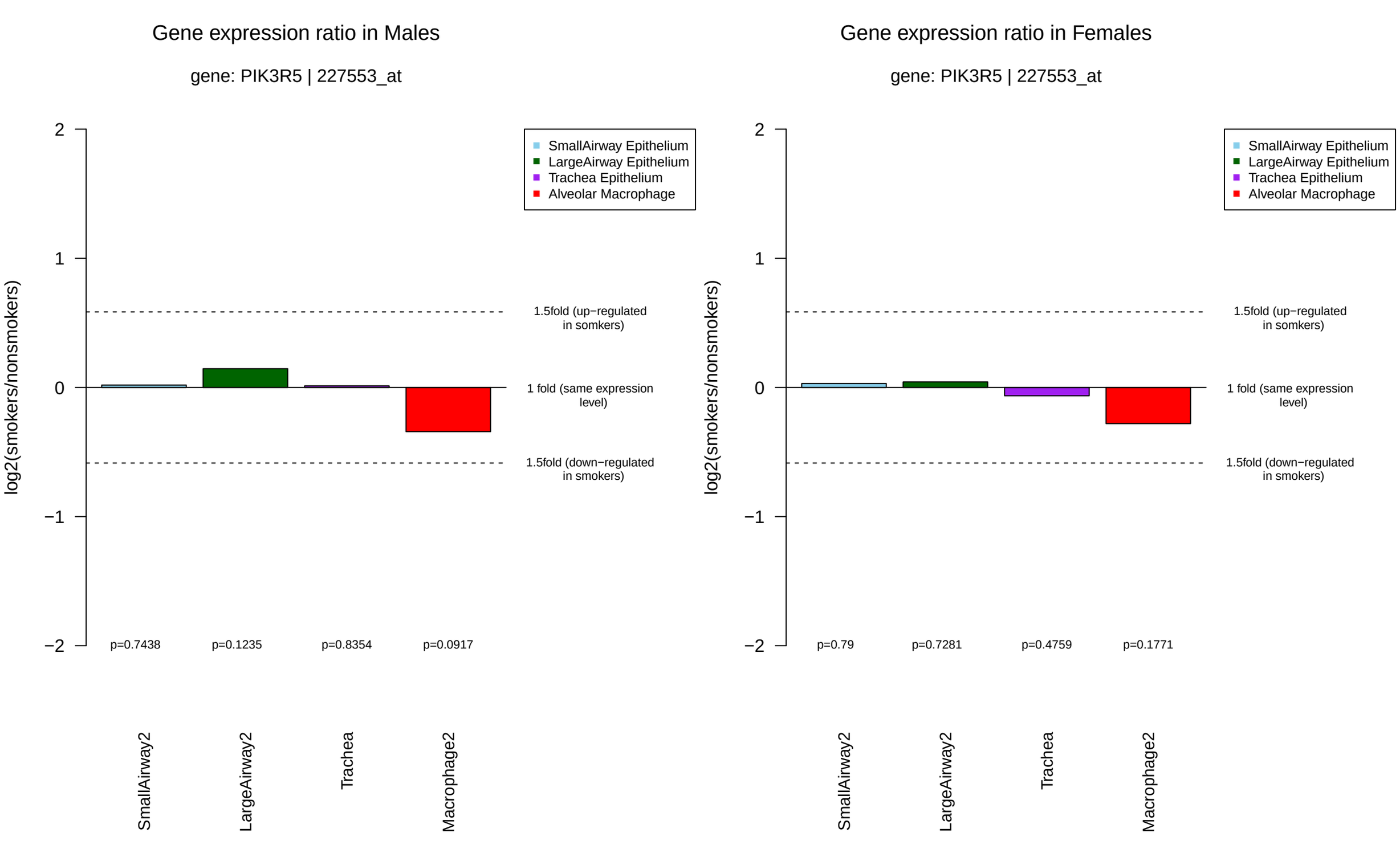
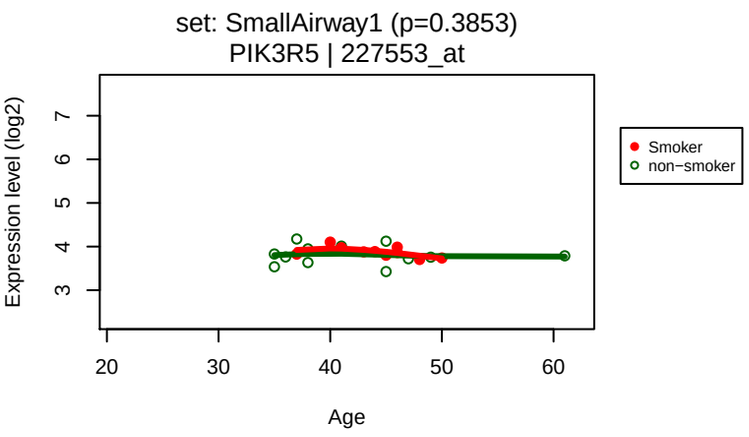
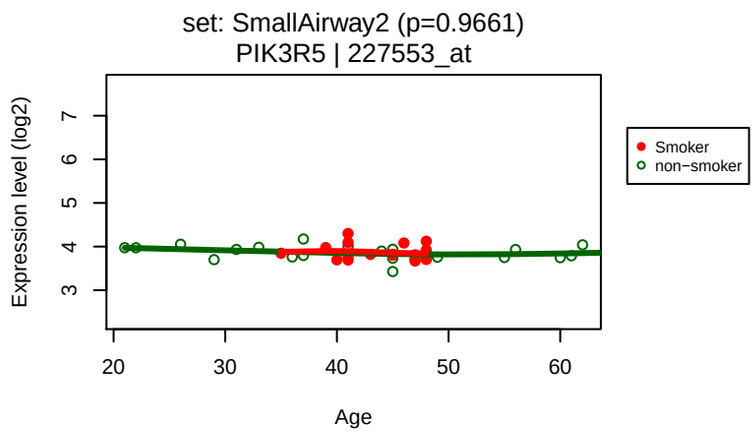
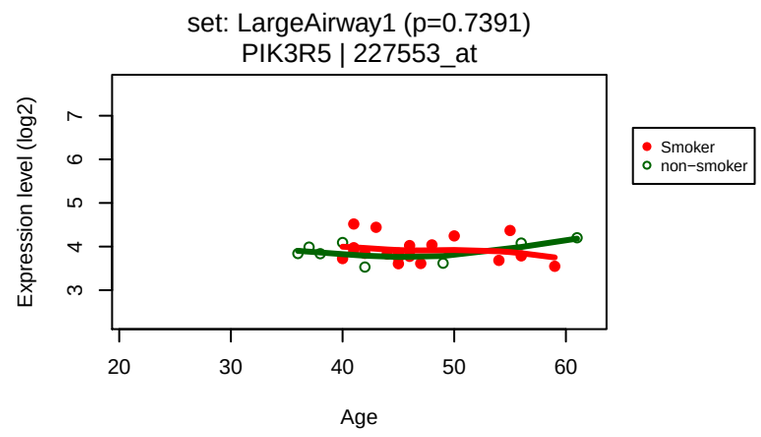

Gene expression of smokers and nonsmokers
gene: PIK3R5 | 227553_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3853
p=0.9661
p=0.3262
p=0.7391
p=0.1585
p=0.8684
p=0.1872
p=0.0309
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

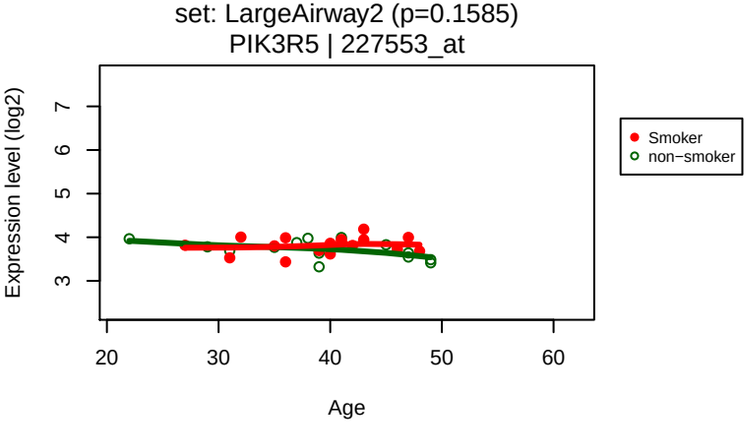

Gene expression ratio (smokers/nonsmokers)
gene: PIK3R5 | 227553_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3853
p=0.9661
p=0.3262
p=0.7391
p=0.1585
p=0.8684
p=0.1872
p=0.0309
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

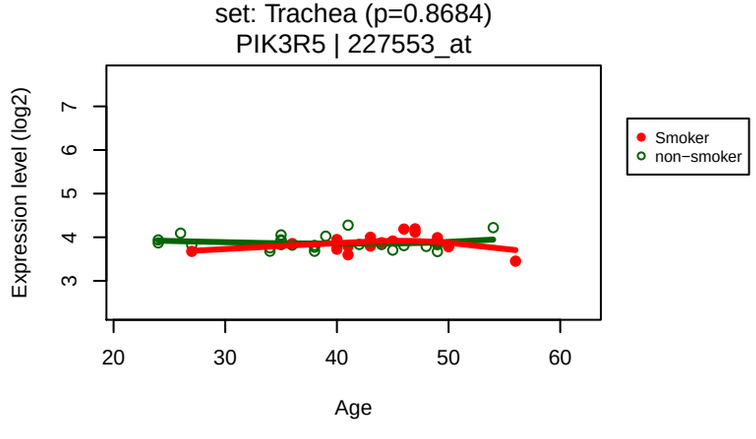

Gene expression ratio in Males
gene: PIK3R5 | 227553_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7438
p=0.1235
p=0.8354
p=0.0917
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIK3R5 | 227553_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.79
p=0.7281
p=0.4759
p=0.1771
SmallAirway2
LargeAirway2
Trachea
Macrophage2

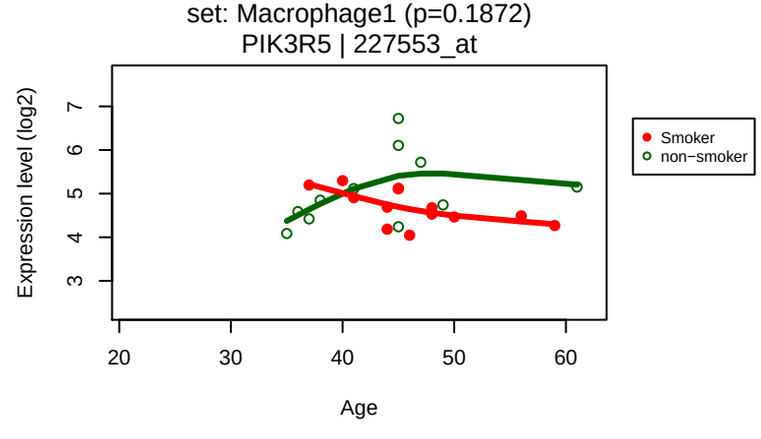

set: SmallAirway1 (p=0.3853)
PIK3R5 | 227553_at
Expression level (log2)
Age
Smoker
non−smoker

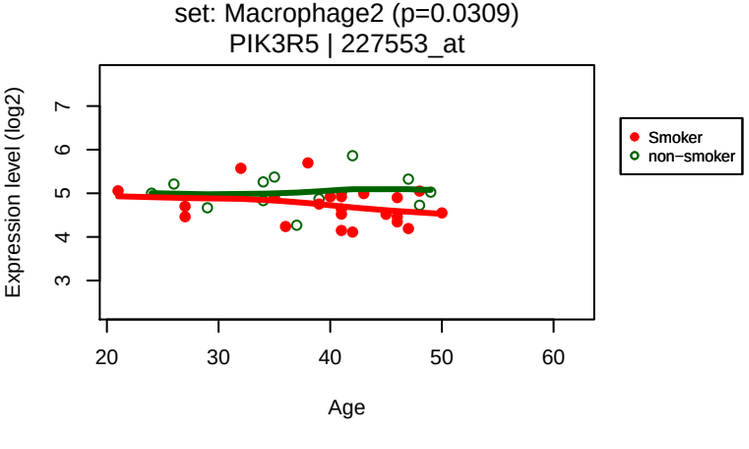

set: SmallAirway2 (p=0.9661)
PIK3R5 | 227553_at
Expression level (log2)
Age
Smoker
non−smoker

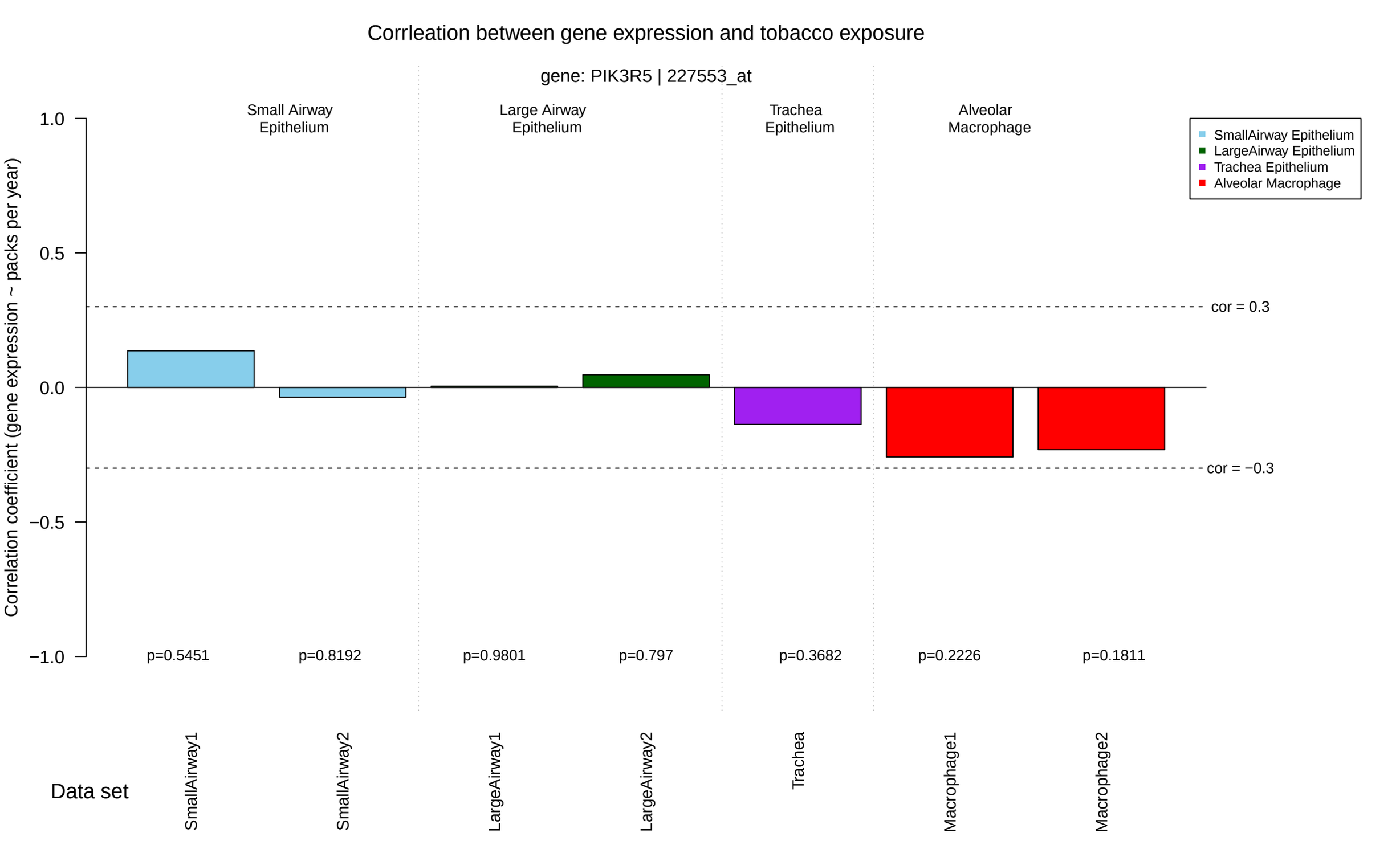

set: LargeAirway1 (p=0.7391)
PIK3R5 | 227553_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1585)
PIK3R5 | 227553_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8684)
PIK3R5 | 227553_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1872)
PIK3R5 | 227553_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0309)
PIK3R5 | 227553_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIK3R5 | 227553_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5451
p=0.8192
p=0.9801
p=0.797
p=0.3682
p=0.2226
p=0.1811
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2